Differentiate exponential functions with base e

1 Differentiate the following:

a $y = e^{x^2+5}$

b $y = e^{x^2+3x+5}$

c $y = e^{4x^2-3x}$

d $y = 3e^{5x} - e^{-4x} + x^2$

e $y = e^{3x} + 5x^2 - 2$

f $y = e^{\frac{1}{6}x^6}$

g $y = 3e^{3x^2}$

h $y = 3e^{x^2-3x+4}$

i $y = 4e^{\sqrt{x}}$

2 Differentiate the following, expressing your answer in factorised form where possible:

a $y = \dfrac{x}{e^x}$

b $y = \dfrac{e^{3x}}{5x}$

c $y = \sqrt{x}e^{-3x}$

d $y = \dfrac{e^x - e^{3x} + 1}{e^x}$

e $y = \dfrac{e^{6x}}{1 + e^x}$

f $y = \dfrac{e^x}{e^{6x} + 7}$

g $y = \dfrac{e^x - 7}{e^x + 7}$

h $y = \dfrac{e^{3x} + 5}{e^{3x} - 5}$

i $y = e^{3x-4}(x + 2)^2$

j $y = e^{-4x}\sqrt{x + 2}$

k $y = \dfrac{e^{2x}}{x}$

l $y = x^5e^{4x}$

m $y = x^5e^{-6x}$

n $y = e^x(x^4 + 1)$

o $y = e^{2x}(x^3 + 4x + 5)$

p $y = (x^2 + 8x + 2)e^{-x}$

q $y = x^3 + x^2e^{3x}$

3 Differentiate the following:

a $y = (e^x - 4)^4$

b $y = (2x - e^{3x})^7$

c $y = (e^{2x} + 5x)^{\frac{3}{4}}$

d $f(x) = (e^x + 4)^5$

e $f(x) = (e^{2x} + 3x)^6$

f $f(x) = (e^{4x} + x^2)^4$

g $f(x) = (5e^{6x} - 9)^7$

h $f(x) = (16 - e^{2x})^3$

i $f(x) = (5x + e^{6x})^7$

j $f(x) = (e^{x^2} - 8)^4$

k $f(x) = \dfrac{1}{(e^x + 1)^6}$

l $f(x) = \dfrac{e^{2x}}{(e^x + 5)^4}$

4 Consider the function $f(x) = e^{3x}(e^x + e^{-x})$.

a Determine $f'(x)$.

b Evaluate $f'(0)$.

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5 Consider the function $f(x) = e^{-3x}\left(x^5 + 6x^2 + 4\right)$. Evaluate $f'(1)$.

Differentiate exponential functions with other bases

6 Consider the function $y = 7^x$.

- **a** Using the fact that $e^{\ln a} = a$, rewrite the function in terms of natural base e.
- **b** Determine y', in terms of the base 7. You may use the substitution $u = (\ln 7)\, x$.
- **c** Hence, determine the exact gradient at $x = 1$.

7 Differentiate the following functions:

- **a** $f(x) = 5^x$
- **b** $f(x) = 2\left(7^x\right)$
- **c** $f(x) = -4^x + 3$
- **d** $f(x) = \dfrac{x^3 - 1}{2^x}$
- **e** $f(x) = x^2\left(5^x\right)$
- **f** $f(x) = 2^{-t}$
- **g** $P(t) = 2^{3t}$
- **h** $f(x) = 5^{2-4x} + 3$

Applications

8 Consider the function $f(x) = e^{2x} + 6x - 8$.

- **a** Find the gradient of the tangent to the curve at $x = 0$.
- **b** Find the equation of the tangent to the curve at $x = 0$.

9 Consider the function $f(x) = \left(e^{2x} - 2\right)^3$.

- **a** Find the gradient of the normal to the curve at $x = 0$.
- **b** Find the equation of the normal to the curve at $x = 0$.

10 Find the equation of the tangent to the curve $y = 2xe^{3x} + x^2 - 8$ at the point $(0, -8)$.

11 Find the equation of the normal to the curve $y = \dfrac{6}{e^{6x} - 4}$ at the point $(0, -2)$.

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